

Premium E-commerce Platform - Backend Architecture Closure Roadmap

Executive Summary

Current architecture maturity is approximately 85%. The project does not require a rewrite. The objective is to close the remaining architectural gaps before continuing large-scale backend implementation.

P0 - Critical Architecture Work

1. InventoryMovement (CREATE)

- New model for stock traceability.
- Movement types: RESERVATION, RELEASE, SALE, RETURN, ADJUSTMENT, RESTOCK.
- Reference support for orders, reservations and manual adjustments.
- Full audit trail.

2. Reservation State Machine (REFACTOR)

- Replace released/confirmed booleans.
- New statuses:
ACTIVE
RELEASED
CONSUMED
EXPIRED
- Centralized transition service.
- Validation of allowed transitions.

3. Order Address Snapshot (IMPLEMENT)

- Preserve historical addresses.
- Snapshot fields stored directly on Order.
- Shipping snapshot fields.
- Billing snapshot fields.
- Existing Address model remains user address book.

P1 - High Value Architecture

4. ProductTranslation

- Translation table for products.
- Localized name.
- Localized description.
- Localized SEO metadata.

5. CategoryTranslation

- Translation table for categories.
- Localized names and SEO fields.

6. StoreSettings

- Currency configuration.
- Default language.
- Checkout settings.
- Inventory settings.

- Tax settings.

7. Event Expansion

- InventoryReservedEvent
- InventoryReleasedEvent
- InventoryConsumedEvent

8. ADR Repository

- Architecture Decision Records.
- Track all major architectural decisions.

9. Domain Documentation

- Per-domain documentation.
- Models.
- Services.
- Business rules.
- Events.

P2 - Engineering Quality

10. Selectors Pattern

- Standardize query access.
- Centralize ORM optimization.

11. Validators Pattern

- Domain validation layer.
- Business rule enforcement.

12. Storage Abstraction

- Local storage implementation.
- S3-ready implementation.
- Storage provider abstraction.

Confirmed Business Rules

Inventory Rules:

- quantity = physical stock.
- reserved_quantity = temporarily blocked stock.
- available_quantity = quantity - reserved_quantity.
- Reservation does NOT reduce quantity.
- Sale confirmation DOES reduce quantity.
- Releasing reservation does NOT modify quantity.

Order Rules:

- OrderItem already stores product snapshots.
- Addresses must be snapshotted on Order.
- Historical orders must remain immutable.

Reservation Rules:

- ACTIVE -> CONSUMED
- ACTIVE -> RELEASED
- ACTIVE -> EXPIRED
- No invalid transitions.

Backend Execution Order

Sprint A

- InventoryMovement
- Reservation State Machine

Sprint B

- Order Address Snapshot

Sprint C

- ProductTranslation
- CategoryTranslation
- StoreSettings

Sprint D

- Event Expansion
- ADR Repository
- Domain Documentation

Sprint E

- Selectors
- Validators
- Storage Abstraction

Final Recommendation

Do not rewrite existing modules.

Keep current Checkout, Payments and Order transition architecture.

Focus on closing the identified architectural gaps.

After completion of P0 and P1 work, Architecture & Business Logic can be considered production-ready and suitable for long-term maintenance and client deployments.